Einstein's general theory of relativity radically departed from all previous thinking about gravitational attraction. Instead of settling for Sir Isaac Newton's view of gravity as spooky action-at-a-distance (a conclusion that made Newton himself uncomfortable), GR regards gravity as the response of a mass to the local curvature of space and time caused by some other mass or field of energy. In other words, concentrations of mass cause distortions—dimples, really—in the fabric of space and time. These distortions guide the moving masses along straight-line geodesics, † † † though they look to us like the curved trajectories we call orbits. The twentieth-century American theoretical physicist John Archibald Wheeler said it best, summing up Einstein's concept as, "Matter tells space how to curve; space tells matter how to move."††††

At the end of the day, general relativity described two kinds of gravity. One is the familiar kind, like the attraction between Earth and a ball thrown into the air, or between the Sun and the planets. It also predicted another variety—a mysterious, anti-gravity pressure associated with the vacuum of space-time itself. Lambda preserved what Einstein and every other physicist of his day had strongly presumed to be true: the status quo of a static universe—an unstable static universe. To invoke an unstable condition as the natural state of a physical system violates scientific credo. You cannot assert that the entire universe is a special case that happens to be balanced forever and ever. Nothing ever seen, measured, or imagined has behaved this way in the history of science, which makes for powerful precedent.

Thirteen years later, in 1929, the American astrophysicist Edwin P. Hubble discovered that the universe is not static. He had found and assembled convincing evidence that the more distant a galaxy, the faster the galaxy recedes from the Milky Way. In other words, the universe is expanding. Now, embarrassed by the cosmological constant, which corresponded to no known force of nature, and by the lost opportunity to have predicted the expanding universe himself, Einstein discarded lambda entirely, calling it his life's "greatest blunder." By yanking lambda from the equation he presumed its value to be zero, such as in this example: Assume $A = B + C$. If you learn later that $A = 10$ and $B = 10$, then A still equals B plus C, except in that case C equals 0 and is rendered unnecessary in the equation.

But that wasn't the end of the story. Off and on over the decades, theorists would extract lambda from the crypt, imagining what their ideas would look like in a universe that had a cosmological constant.